

Remote Access Tools

Sub-techniques (3) 

An adversary may use legitimate remote access tools to establish an interactive command and control channel within a network. Remote access tools create a session between two trusted hosts through a graphical interface, a command line interaction, a protocol tunnel via development or management software, or hardware-level access such as KVM (Keyboard, Video, Mouse) over IP solutions. Desktop support software (usually graphical interface) and remote management software (typically command line interface) allow a user to control a computer remotely as if they are a local user inheriting the user or software permissions. This software is commonly used for troubleshooting, software installation, and system management.^{[1][2][3]} Adversaries may similarly abuse response features included in EDR and other defensive tools that enable remote access.

Remote access tools may be installed and used post-compromise as an alternate communications channel for redundant access or to establish an interactive remote desktop session with the target system. It may also be used as a malware component to establish a reverse connection or back-connect to a service or adversary-controlled system.

Installation of many remote access tools may also include persistence (e.g., the software's installation routine creates a [Windows Service](#)). Remote access modules/features may also exist as part of otherwise existing software (e.g., Google Chrome's Remote Desktop).^{[4][5]}

ID: T1219

Sub-techniques: [T1219.001](#), [T1219.002](#), [T1219.003](#)

① **Tactic:** [Command and Control](#)

① **Platforms:** Linux, Windows, macOS

Contributors: Dray Agha, @Purp1eW0lf, Huntress Labs; Matt Kelly, @breakersall; Zachary Stanford, @svch0st

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Procedure Examples

ID	Name	Description
G1024	Akira	Akira uses legitimate utilities such as AnyDesk and PuTTY for maintaining remote access to victim environments. ^{[6][7]}
G1043	BlackByte	BlackByte has used tools such as AnyDesk in victim environments. ^{[8][9]}
S0030	Carbanak	Carbanak has a plugin for VNC and Ammyy Admin Tool. ^[10]
G0008	Carbanak	Carbanak used legitimate programs such as AmmyyAdmin and Team Viewer for remote interactive C2 to target systems. ^[11]
G0080	Cobalt Group	Cobalt Group used the Ammyy Admin tool as well as TeamViewer for remote access, including to preserve remote access if a Cobalt Strike module was lost. ^{[12][13][14]}
G0105	DarkVishnya	DarkVishnya used DameWare Mini Remote Control for lateral movement. ^[15]
S0384	Dridex	Dridex contains a module for VNC. ^[16]
S0554	Egregor	Egregor has checked for the LogMein event log in an attempt to encrypt files in remote machines. ^[17]
G0046	FIN7	FIN7 has utilized the remote management tool Atera to download malware to a compromised system. ^[18]
G0115	GOLD SOUTHFIELD	GOLD SOUTHFIELD has used the cloud-based remote management and monitoring tool "ConnectWise Control" to deploy REvil . ^[19]
S0601	Hildegard	Hildegard has established tmate sessions for C2 communications. ^[20]
G1032	INC Ransom	INC Ransom has used AnyDesk and PuTTY on compromised systems. ^{[21][22][23][24]}
S1245	InvisibleFerret	InvisibleFerret has utilized remote access software including AnyDesk client through the "adc" module. ^{[25][26][27]} InvisibleFerret has also downloaded the AnyDesk client should it not already exist on the compromised host by searching for <code>C:/Program Files (x86)/AnyDesk/AnyDesk.exe</code> . ^[28]
G1051	Medusa Group	Medusa Group has leveraged Remote Access Software for lateral movement and data exfiltration. ^{[29][30][31][32]} Medusa Group has also been known to utilize Remote Access Software such as AnyDesk, Atera, ConnectWise, eHorus, N-Able, PDQ Deploy, PDQ Inventory, SimpleHelp and Splashtop. ^[30]
C0002	Night Dragon	During Night Dragon , threat actors used several remote administration tools as persistent infiltration channels. ^[33]
G0049	OilRig	OilRig has incorporated remote monitoring and management (RMM) tools into their operations including ngrok . ^[34]
C0060	Operation AkaiRyū	During Operation AkaiRyū , MirrorFace used remote access tools including PuTTY. ^[35]
S0148	RTM	RTM has the capability to download a VNC module from command and control (C2). ^[36]
G0034	Sandworm Team	Sandworm Team has used remote administration tools or remote industrial control system client software for execution and to maliciously release electricity breakers. ^{[37][38]}
G1057	ShinyHunters	ShinyHunters has used MeshCentral and ConnectWise to gain initial access, to run administrative command queries and to deploy the custom lateral movement and defacement script [victim_abbreviation]_fanout.sh. ^{[39][40]}
G0139	TeamTNT	TeamTNT has established tmate sessions for C2 communications. ^{[20][41]}

Mitigations

ID	Mitigation	Description
M1042	Disable or Remove Feature or Program	Consider disabling unnecessary remote connection functionality, including both unapproved software installations and specific features built into supported applications.
M1038	Execution Prevention	Use application control to mitigate installation and use of unapproved software that can be used for remote access.
M1037	Filter Network Traffic	Properly configure firewalls, application firewalls, and proxies to limit outgoing traffic to sites and services used by remote access software.
M1034	Limit Hardware Installation	Block the use of IP-based KVM devices within the network if they are not required.
M1031	Network Intrusion Prevention	Network intrusion detection and prevention systems that use network signatures may be able to prevent traffic to remote access services.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0496	Behavior-Chain Detection for Remote Access Tools (Tool-Agnostic)	AN1366	Chain of remote access tool behavior: (1) initial execution of remote-control/assist agent or GUI under user context; (2) persistence via service or autorun; (3) long-lived outbound connection/tunnel to external infrastructure; (4) interactive control signals such as shell or file-manager child processes spawned by the RAT parent.
		AN1367	Sequence of RAT agent execution, systemd persistence, and long-lived external egress; optional interactive shells spawned from the agent.
		AN1368	Electron/GUI or headless RAT execution followed by LaunchAgent/Daemon persistence and persistent external connections; interactive children (osascript/sh/curl) spawned by parent.

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